THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Why is it risky if many people in a community skip vaccination? 1

(a) Water becomes impure
(b) Vaccines cause the disease
(c) The pathogen can spread through unvaccinated people
(d) Antibodies disappear

2. A student winds 50 turns on a nail and lifts 8 pins, and 100 turns and lifts 17 pins. The data show that: 1

(a) pins have no mass **(b)** no relation
(c) strength falls with turns **(d)** strength rises with turns

3. A student heats three metal rods of equal size and finds wax dropping at different times. The dependent variable is: 1

(a) time for wax to drop **(b)** metal type
(c) length of rod **(d)** flame

4. Which is NOT a feature of a unicellular organism? 1

(a) cell membrane **(b)** tissues
(c) nucleus (in protozoa) **(d)** cytoplasm

5. A new disease spreads from person to person. Which first step reduces the spread? 1

(a) closing hospitals **(b)** stopping all food
(c) isolating the patients and hygiene **(d)** ignoring symptoms

6. A student's 'magnet' was a steel rod in a coil and kept its magnetism after the current stopped. This shows that steel: 1

(a) retains magnetism **(b)** cannot be magnetised
(c) is an insulator **(d)** loses magnetism quickly

7. Assertion (A): Records should include observations that do not match the expectation. 1
Reason (R): Selective reporting gives an incomplete and misleading picture.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): A red blood cell can carry more oxygen because it has no nucleus. Reason (R): Haemoglobin takes up the space gained. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): Diabetes is a communicable disease. Reason (R): Diabetes spreads from person to person by contact. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): A compass needle placed under a current-carrying wire deflects. Reason (R): Electric current produces a magnetic field around the wire. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. A pendulum completes 30 swings in 48 s. The time period is: 1

(a) 16 s **(b)** 78 s **(c)** 1.6 s **(d)** 0.625 s

12. Which sequence correctly lists in increasing size: red blood cell (7 micrometres), bacterium (2 micrometres), virus (0.1 micrometre)? 1

(a) red blood cell, bacterium, virus **(b)** virus, bacterium, red blood cell
(c) bacterium, virus, red blood cell **(d)** virus, red blood cell, bacterium

13. A sick person drinks water from a well that is next to a leaking toilet pit. The disease most likely to spread is: 1

(a) rickets **(b)** goitre **(c)** typhoid **(d)** scurvy

14. A device connected to a 5 A fuse draws 3 A and a second device draws 4 A. If both are on in the same circuit, the fuse will: 1

(a) melt **(b)** become magnetic **(c)** remain intact **(d)** cool

15. Which is the best question to start a study of how the volume of water affects the time to boil? 1

(a) Who boils water?
(b) Is boiled water good?
(c) Does 200 mL of water boil faster than 400 mL on the same burner?
(d) Where do we boil water?

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are public toilets and clean drains important for community health? 2

17. An electric cell goes weak. What happens to the strength of an electromagnet connected to it? 2

18. The mean of five readings is 12.4 s. Four readings are 12.0, 12.6, 12.5 and 12.3. Find the fifth. 2

19. Why did Hooke see only walls in cork cells and not the contents? 2

20. Why is a person more likely to recover quickly from the flu with rest and fluids rather than antibiotics? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Compare the effects of a vaccine and of a medicine on a person already ill with measles. 3

22. A bulb draws 0.3 A and a heater draws 7 A from the same supply. Which needs thicker wiring? Give a reason. 3

23. Explain how bias can affect an investigation and how to reduce it. 3

24. Explain what would happen to a multicellular organism if its cells did not specialise. 3

OR

Explain how mosquito-borne diseases can be controlled by an individual, the family and the community. 3

25. A teacher passes current through a wire wrapped in paper and a wire wrapped in iron foil near a compass. Explain the expected observations. 3

OR

Differentiate between an experiment and a survey with an example each. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A student's results for plant height (cm) after 4 weeks with watering of 0, 20, 40, 60, 80 mL were 3, 8, 12, 13, 9. Describe the pattern, suggest an explanation and a next experiment. 5

OR

A cell is magnified 800x using a 10x eyepiece. Find the objective power. If the image is 4 mm, find the actual size in micrometres. 5

27. Explain the working of vaccines with a graph-like description of antibody levels after the first and the second dose, and say why boosters are needed. 5

OR

A heater draws 4.5 A, a TV 0.5 A and a bulb 0.3 A. Find the total current, say if a 5 A fuse is suitable and explain what happens if it is used. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND